Minjee Kim

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Sufficient dimension reduction (SDR) for complex data structures, with applications in biomedical statistics

REFERENCES

Jae Keun Yoo (Advisor)
Professor, *Department of Statistics, Ewha Womans University*
peter.yoo@ewha.ac.kr

Kyongwon Kim (Corresponding Author)
Assistant Professor, *Department of Applied Statistics and Department of Statistics and Data Science, Yonsei University*
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Michael Hardisty (Corresponding Author)
Scientist, *Holland Bone and Joint Program, Sunnybrook Research Institute*
Assistant Professor, *Department of Surgery, University of Toronto*
michael.hardisty@sunnybrook.ca

EDUCATION

Sep 2022 – Feb 2027 **Ewha Womans University**
- *Ph.D.* in Statistics (Advisor: **Jae Keun Yoo**)
  All academic requirements completed; degree to be conferred in February 2027
  T.A.: Regression Analysis, Bayesian Statistics, Mathematical Statistics, etc.

Mar 2017 – Aug 2022
- *B.S.* in Architectural and Urban Systems Engineering / Statistics

Jan 2025 – Jun 2025 **University of Toronto**
- *Visiting Student* in Mechanical & Industrial Engineering

RESEARCH EXPERIENCES

Sep 2022 – Feb 2027 **Ewha Dimension Reduction and Intelligent Learning Lab**
Graduate Student Researcher
- Extended SDR methods to functional data, survival regression, non-elliptical settings

Jan 2025 – Jun 2025 **Sunnybrook Health Sciences Centre**
Visiting Researcher
- Developed a dynamic mortality prediction model for prostate cancer patients

PUBLICATIONS

1. **Non-elliptical Dimension Reduction in Survival Regression**,
   Minjee Kim, Minjeong Kim and Jae Keun Yoo, 2026, *Stat*, 15(3), e70162.
2. **Dynamic Mortality Prediction for Prostate Cancer based on Irregularly Sampled Multimodal EHR and Musculoskeletal Image-based Biomarkers**,
   Eunseon Seong, Minjee Kim, Dareen Eom, Gyeonghwa Heo, Urban Emmenegger, Saleh Tabatabaei, Tayler Declan Ross, and Michael Hardisty, 2026, In *Pacific-Asia Conference on Knowledge Discovery and Data Mining* (pp. 399–410).
3. **Nonlinear Functional Sufficient Dimension Reduction via Principal Fitted Components**,
   Minjee Kim, Yujin Park, Kyongwon Kim, and Jae Keun Yoo, 2025, *Statistics and Computing*, 35(4), 95.
4. **Covariance Projective Resampling Informative Predictor Subspace for Multivariate Regression**,
   Minjee Kim and Jae Keun Yoo, 2025, *Journal of the Korean Statistical Society,* 54(3), 920–931.
5. **Applications of response dimension reduction in large *p*-small *n* problems**,
   Minjee Kim and Jae Keun Yoo. 2024, *Communications for Statistical Applications and Methods*, 31(2), 191–202.

IN PREPARATION

1. **Second-Order Covariance Projective Resampling for Sufficient Dimension Reduction with Multivariate Responses,** Minjee Kim and Jae Keun Yoo. *Statistics and Computing, Under review.*
2. **Projective-Resampling Sliced Mean Regression,** Minjee Kim and Jae Keun Yoo. *Work in progress.*

CONFERENCES

Jun 2026	**30th PAKDD** – "Dynamic Mortality Prediction for Prostate Cancer using Multimodal EHR"
Dec 2025	**13th IASC-ARS** – "covPRIPS for Multivariate Regression" (**Session chair**)
Aug 2025	**8th EcoSta** – "Nonlinear Functional SDR via Principal Fitted Components"
Dec 2024	**IASC-ARS Interim** – "Response dimension reduction in large *p*-small *n* problems"

SKILLS

- **R** (primary), **Python / SQL** (clinical EHR data querying and preprocessing)
- Languages – English (academic), Korean (native)

HONORS & SERVICE

Jul 2026	*Volunteer*, 33rd International Biometric Conference
Aug 2025	Ewha Womans University *Best Paper Award*
Nov 2022, Apr 2023	*Workshop Assistant,* Ewha GSIS Computational Social Science Workshop